



Fraud detection that doesn't guess. It reads the web.

An **autonomous AI agent** that scores every transaction, reads the same public web a human investigator would — via the **Bright Data SERP API** — explains its verdict in plain English, and **freezes accounts** the instant risk goes HIGH.

AI Agent

Bright Data SERP API

FastAPI

Next.js

DeepSeek LLM

Real-Time Risk

Fraud engines reason in **a vacuum**.

A traditional engine sees **amount**, **user history**, maybe a velocity rule — and nothing about *who the merchant actually is*.

A brand-new shell merchant called **"FreeMoneyCryptoLottery"** looks identical to a legitimate one to a model that only sees numbers.

The signal that would expose it — **scam reports, Trustpilot ratings, BBB complaints, Reddit r/scams threads, news coverage** — lives on the **open web**: behind rate limits, bot detection, and geo-blocks that internal systems were never built to reach.

→ Exactly the wall this hackathon targets: agents

**throttled by stale,
locked data**

A NUMBERS-ONLY ENGINE SEES

- ▶ amount = \$9,000
- ▶ user avg = \$45
- ▶ location = a string

...BUT IS BLIND TO

- ▶ **Who** the merchant is
- ▶ scamadviser / ripoffreport hits
- ▶ r/scams threads · 1.2★ Trustpilot

Live web context on **every transaction.**

Five layers, combined by **maximum risk** — so no layer can ever silently weaken the verdict.

1

Deterministic risk floor

amount · velocity · impossible-travel geo · structuring

2

Bright Data SERP API → merchant web reputation

reads scam reports, Trustpilot, BBB, r/scams, news — scored 0–100

3

DeepSeek LLM reasoning

plain-English, compliance-grade explanation over the real evidence

4

Combine by MAX risk

no layer can be silently overridden downward

5

HIGH → autonomous account freeze

rejects every further transaction (423 Locked) until a human unfreezes

Bright Data SERP API — the heart of the agent.

One Google search per merchant, returned as structured JSON, scored deterministically into a 0–100 reputation.

```
# one search per merchant
query = "{merchant} reviews scam complaints"
POST https://api.brightdata.com/request
{
  "zone": "serp_api1",
  "url": ".../search?q=<q>&brd_json=1",
  "format": "raw"
}
Authorization: Bearer $BRIGHTDATA_API_KEY
```

brd_json=1 returns the SERP as JSON. Sentinel parses the top 5 organic results and scores them — base **50**, clamped to **[0,100]**.

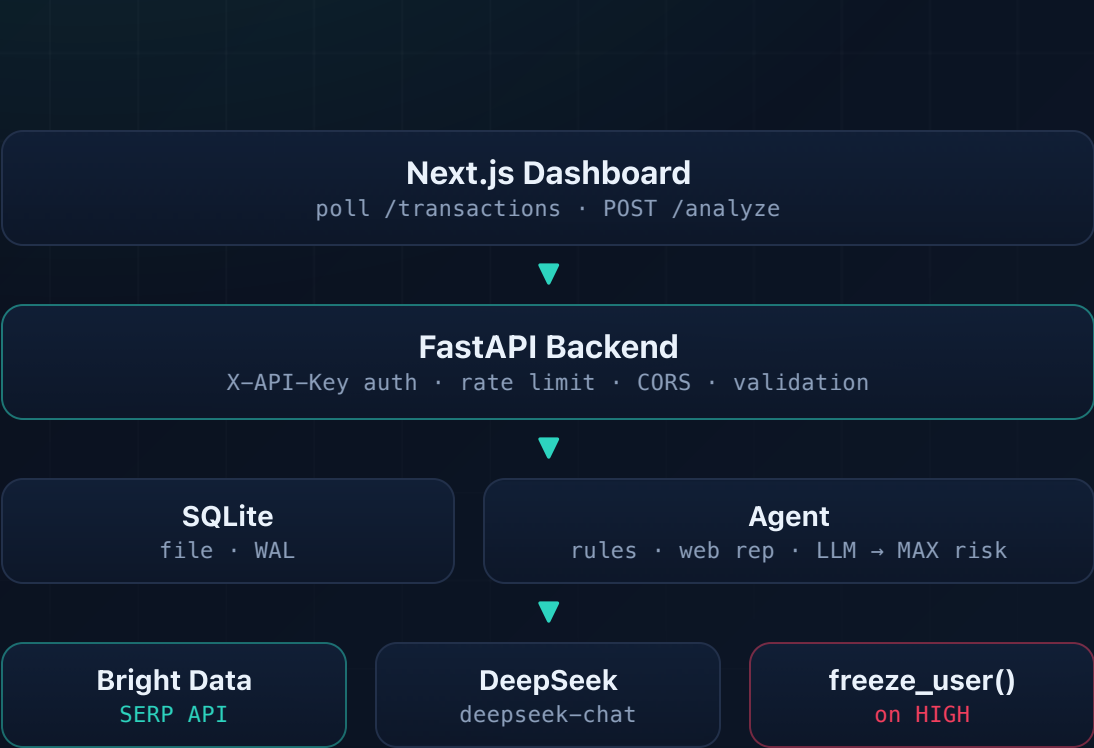
WEB SIGNAL IN THE SERP	SCORE
Scam-flag domain (scamadviser, ripoffreport...)	-30 ea
Trustpilot rating ≥ 4.5★	+20
Trustpilot rating ≤ 2.0★	-25
BBB (bbb.org) listing	+15
Reddit r/scams hit	-20
Reputable news (Reuters, BBC, WSJ...)	+10

WIRED INTO THE DECISION — NOT COSMETIC

score < **30** → ≥ **MEDIUM**, < **15** → **HIGH** · cached **24h** (1st sighting live ~1–3s, then sub-ms) · **disabled-safe** if no key.

Defense in depth, one decision.

A FastAPI agent fuses deterministic rules, live web reputation, and LLM reasoning — then acts.



LAYER	WHAT IT CATCHES
Amount / ratio	≥ \$5k or ≥ 30× avg → HIGH
Velocity	≥ 5 txn / 5 min → HIGH
Geo	impossible travel < 5 min → HIGH
Structuring	rolling 60-min ≥ \$20k → MEDIUM
Web rep floor	score < 30 → ≥ MEDIUM; < 15 → HIGH
LLM reasoning	narrative — can only <i>raise</i> the floor

Dual-mode: real key → **DeepSeek**; no key → fully **deterministic heuristic**, so the demo always works.

Live web → cache, in real time.

Every merchant verdict is auditable — and the second sighting is sub-millisecond.

Sentinel — Live Transactions

LIVE

\$42.00	Blue Bottle Coffee New York, NY ·  live lookup → resolved	LOW	92
\$48.00	Blue Bottle Coffee New York, NY · cached · c (sub-ms)	LOW	92
\$9,000	Unknown Wire Transfer Lagos, Nigeria ·  live lookup → scam hits	HIGH	12

 ACCOUNT FROZEN — autonomous · further transactions rejected (423)

- ▶ New merchant row shows  **live web lookup** → resolves to a **green** badge (~1–3s).
- ▶ Next sighting renders instantly with a **·c cached** marker — the live-web → cache beat.
- ▶ Bad merchant (**Unknown Wire Transfer**) → **red** badge → risk **HIGH** → **account frozen**.
- ▶ Click the badge → modal shows the **actual SERP evidence**: scamadviser, ripoffreport, r/scams — clickable.
- ▶ Retry that user → **rejected (423)**, proving the autonomous freeze holds.

Production-shaped, **not a toy.**

Built for the Security & Compliance track — third-party / merchant risk, investigated autonomously.

WHY IT MATTERS

- ▶ **Autonomous action** — agent investigates risk across the open web and freezes accounts with no human in the loop.
- ▶ **Merchant / third-party risk** — the exact gap traditional engines miss.
- ▶ **Production-ready** — API-key auth, locked CORS, 60/min rate limit, strict validation, 423 freeze enforcement.
- ▶ **Graceful degradation** — every external call (Bright Data, DeepSeek) has a defined failure mode that never blocks a transaction.

WHAT'S NEXT

- ▶ **Web Unlocker / Scraping Browser** — read full scam-report pages, not just SERP snippets.
- ▶ **Continuous re-scoring** of known merchants beyond the 24h TTL.
- ▶ **Sanctions / watchlist screening** via live web sources — deepens the compliance angle.
- ▶ **Per-category risk policies** — tunable thresholds by merchant type.

LINKS

GitHub: github.com/<you>/sentinel

Demo: <http://localhost:3000> · API <http://localhost:8000>

Web data: Bright Data SERP API